2.5D Topology Optimization

2.5D | Plane Stress | Optimization | FEA Engine

Minimum-thickness method for 2.5D topology optimization applied to structural design.

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INTRODUCTION

Objective

This app helps users generate an efficient 2.5D membrane layout with variable thickness for structures subjected to in-plane loads. Its goal is to distribute a limited amount of material in the domain so that the final design becomes as stiff as possible. In optimization terms, this is posed as a compliance minimization problem: the lower the compliance, the stiffer the structure. The method is based on a 2.5D topology optimization approach, where the design variable is the element thickness rather than a conventional density variable. This allows the optimizer to place more material in critical regions without forcing unrealistic in-plane member dimensions.

Limitations

This tool is intended for plane-stress membrane problems with dominant in-plane structural behavior. The underlying formulation assumes that the out-of-plane thickness remains small compared with the in-plane dimensions, which is why the method is described as 2.5D rather than fully 3D. It is not intended for problems where torsion, out-of-plane flexure, or other complex 3D effects control the response. The reference method is also developed within a linear elastic, small-displacement framework.

Optimization workflow in the app

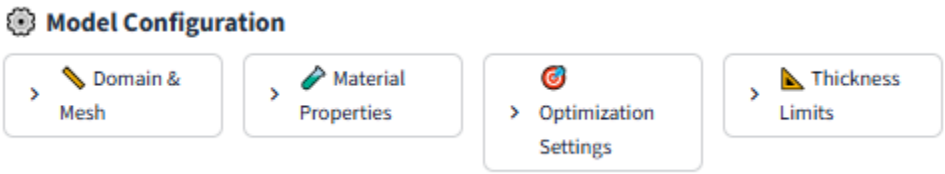
From the user's perspective, the process is straightforward:

1. Define the geometry and mesh.
2. Assign material properties.
3. Enter supports and loads.
4. Set the volume fraction, filter radius, iteration limit, and thickness bounds.
5. Run the optimization.
6. Review the final thickness map, inspect the evolution history, and explore the 3D result.
7. Export the final geometry if needed.

This workflow translates the methodology of the reference paper into an interactive design tool for fast conceptual exploration.

INPUTS

The app is organized so the user defines the structural model, optimization parameters, and loading conditions before running the solver.



Domain and mesh

These inputs determine the geometric resolution of the optimization.

- Domain X (m) and Domain Y (m) define the size of the rectangular design region.
- Mesh Size (m) controls the finite element discretization. A finer mesh captures more detail, while a coarser mesh runs faster.

Material properties

- Elastic Modulus, E (Pa) controls structural stiffness.
- Poisson's Ratio, ν defines the constitutive behavior of the membrane in plane stress.
- Material Density, ρ is used when self-weight is included.
- Include Self-Weight adds gravity loading to the model.

Optimization settings

- Volume Fraction sets the amount of material allowed in the final design.
- Filter Radius, rmin regularizes the optimization and helps control numerical artifacts such as checkerboard patterns and mesh dependency, which are common concerns in topology optimization.
- Max Iterations sets the maximum number of optimization steps.

Thickness limits

These two inputs are central to the method because the design evolves through a variable thickness field. The reference paper shows that allowing a range of thicknesses improves the structural response and produces clearer, more practical member layouts than fixed-thickness 2D density approaches.

- Min Thickness, tmin is the lower thickness bound.
- Max Thickness, tmax is the upper thickness bound.

Model Configuration

Domain & Mesh

Domain X (m)

4,00 − +

Domain Y (m)

24,00 − +

Mesh Size (m)

0,050 − +

Grid: 80 x 480 elements

Total: 38,400

Material Properties

Elastic Modulus (Pa)

15000 − +

Poisson's Ratio (ν)

0.30

Material Density (ρ)

2,400 − +

☐ Include Self-Weight

Optimization Settings

Volume Fraction

0.50

Filter Radius (rmin)

0,20 − +

Max Iterations

50 − +

Thickness Limits

Min Thickness (m)

0,1000 − +

Max Thickness (m)

1,0000 − +

SETUP

Supports

Can be entered interactively or edited in table form:

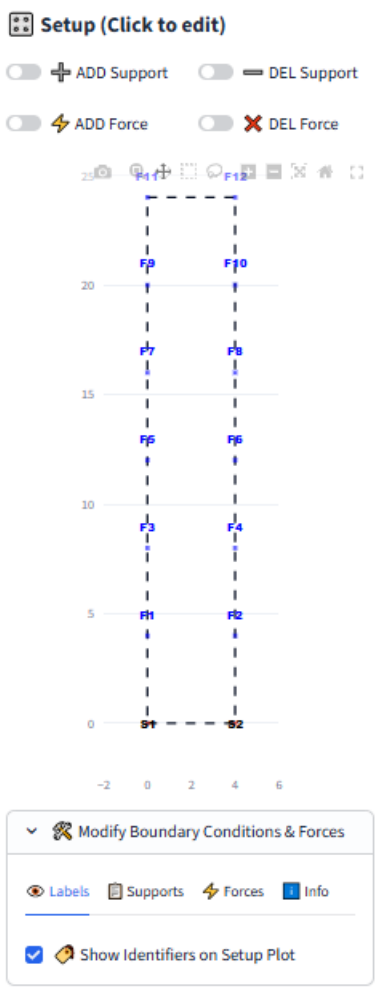
- Fixed restrains displacement in both X and Y.
- Pinned X restrains X displacement only.
- Pinned Y restrains Y displacement only.

Loads

Are applied as localized in-plane point forces:

- Fx (N) is the horizontal component.
- Fy (N) is the vertical component.

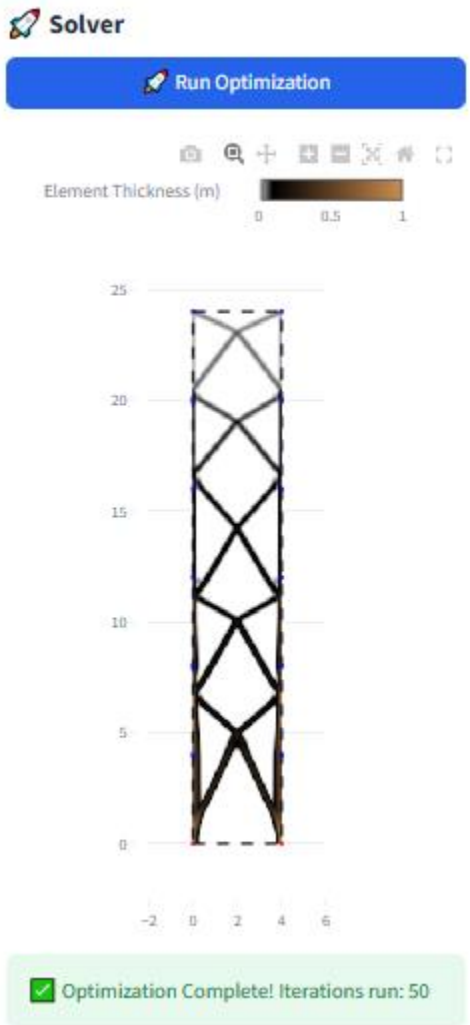
Negative values indicate action in the negative global direction.



OUTPUTS

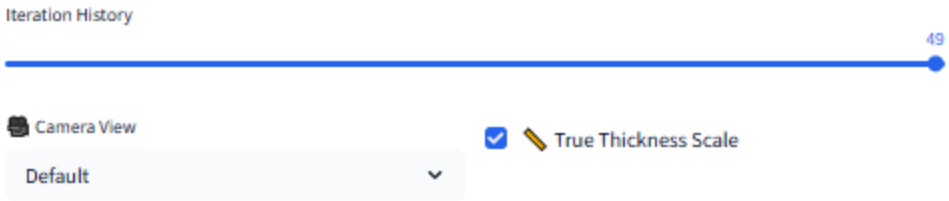
Optimized 2D thickness map

The primary output is a 2D color map of thickness over the design domain. Thicker regions identify the main load-carrying paths, while regions near the minimum thickness indicate areas of low structural demand.



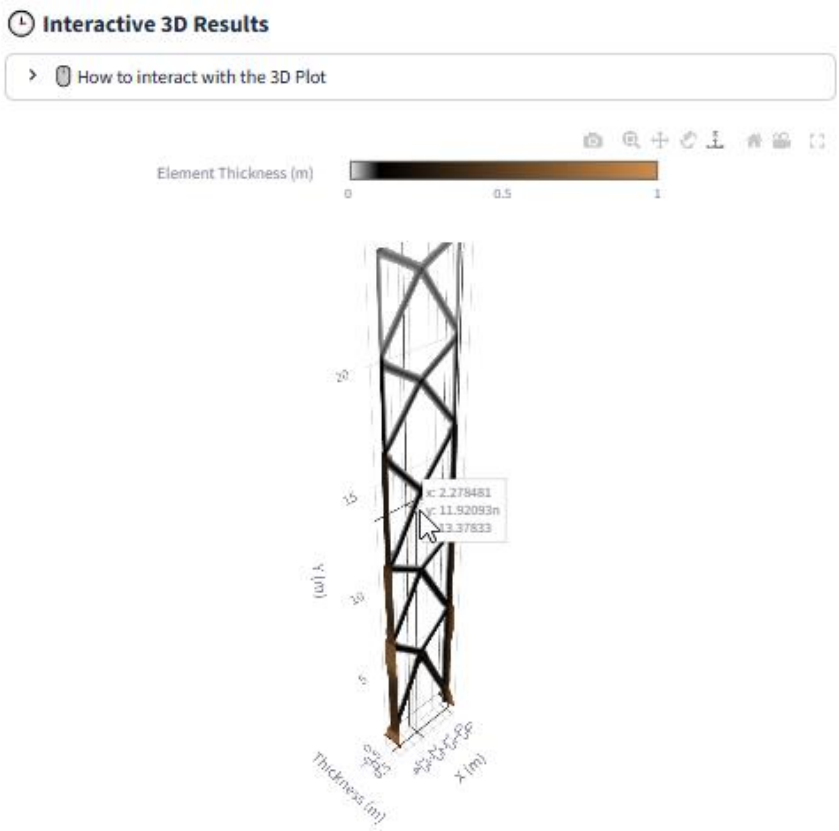
Iteration history and View options

The app records the optimization history, allowing the user to inspect how the thickness distribution evolves from the initial state to the final solution. As well as predefined view options to picture the design.



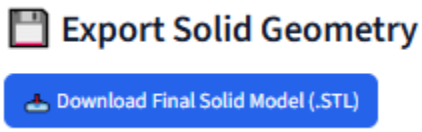
Interactive 3D visualization

The optimized 2D thickness field is extruded into a 3D solid representation. This helps users better interpret the result as a variable-thickness structural form.



STL export

The final solution can be exported as an STL file, which is useful for presentation models, geometric post-processing, or fabrication-oriented workflows.



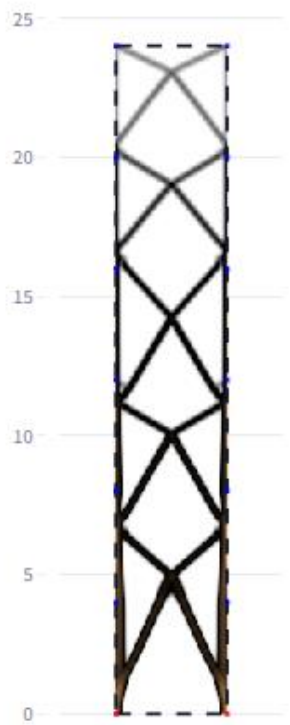
HOW TO USE THE RESULTS

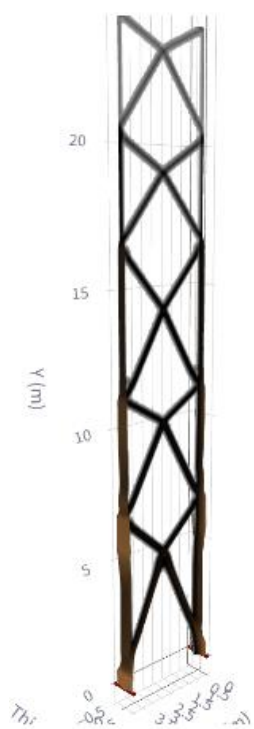
The outputs are best interpreted as a conceptual structural design guide. They help identify:

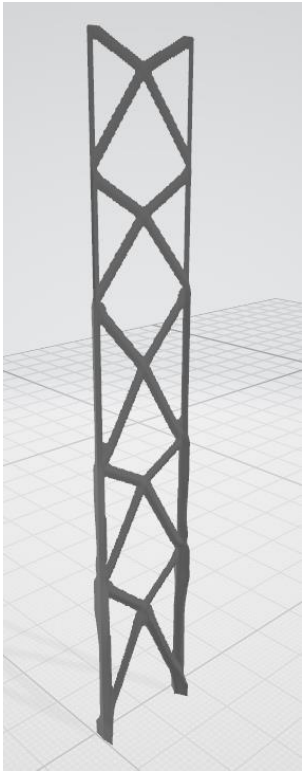
- the main load paths,
- zones that require more material,
- regions that can remain thin,
- and the overall structural layout that emerges from the optimization.

The reference paper emphasizes that this formulation tends to produce structural members that are easier to identify and that have more reasonable in-plane proportions than conventional 2D density solutions.

Thickness map **3D Preview** **STL**







METHODOLOGY

The app follows the minimum-thickness method for 2.5D topology optimization proposed by Pozo et al. The central idea is simple but powerful: instead of assigning each element a fictitious density between 0 and 1, the method assigns each element a physical thickness variable. This makes the design easier to interpret structurally and allows the optimizer to distribute material through thickness in a direct and meaningful way.

Design variable

The design is represented by the vector

$$\mathrm{z} = [z_1, z_2, \dots, z_{N_e}]$$

where $z_i$is the thickness assigned to element i. Each thickness is bounded by lower and upper limits:

$$z_{\min} \leq z_i \leq z_{\max}$$

This is one of the defining differences of the method. In traditional density optimization, the variable controls stiffness indirectly through a material interpolation law. Here, the variable is the actual thickness of the membrane element.

Optimization problem

The reference formulation uses a compliance minimization problem with volume and equilibrium constraints:

$$\min_{\mathbf{z}} \boldsymbol{J}(\mathbf{z}) = \mathbf{f}^{\boldsymbol{T}}\mathbf{u}(\mathbf{z})$$

subject to

$$\boldsymbol{V}(\mathbf{z}) - \boldsymbol{V_{obj}} \leq \mathbf{0}$$
$$\mathbf{f} - \mathbf{K}(\mathbf{z})\mathbf{u}(\mathbf{z}) = \mathbf{0}$$
$$\boldsymbol{z}_{\mathbf{min}} \leq \boldsymbol{z} \leq \boldsymbol{z}_{\mathbf{max}}$$

where:

- $\mathbf{f}$is the nodal force vector,
- $\mathbf{u}$is the displacement vector,
- $\mathbf{K}$is the global stiffness matrix,
- $\boldsymbol{V}(\mathbf{z})$is the total structural volume,
- and $\boldsymbol{V_{obj}}$is the prescribed target volume.

Because compliance is a measure of flexibility, minimizing it makes the structure stiffer.

Volume constraint

The total volume is computed by summing the volume of every element:

$$\boldsymbol{V}(\mathbf{z}) = \sum_{i=1}^{N_e} \boldsymbol{V}_i \quad \text{with} \quad \boldsymbol{V}_i = \boldsymbol{z}_i \boldsymbol{A}_i$$

where $\boldsymbol{A}_i$is the in-plane area of element $\boldsymbol{i}$. This means the material constraint is enforced directly through the thickness field.

Why a penalized thickness is needed

A pure thickness optimization approach can produce solutions that look like plates with large regions of almost-zero thickness. While mathematically possible, those results are less useful in structural design because they do not clearly define voids or well-formed members. To address this, the paper introduces a penalization on stiffness based on a reference thickness $\boldsymbol{z}_{ref}$. Elements below this threshold lose stiffness efficiency, which encourages the optimizer to either keep them structurally meaningful or allow them to become effectively void-like.

The ideal penalized thickness is written as

$$\boldsymbol{z}_i^*(\boldsymbol{z}_i) = \begin{cases} \boldsymbol{z}_{\min}, & \boldsymbol{z}_i < \boldsymbol{z}_{ref} \\ \boldsymbol{z}_i, & \boldsymbol{z}_i \geq \boldsymbol{z}_{ref} \end{cases}$$

with derivative

$$\frac{d\boldsymbol{z}_i^*}{d\boldsymbol{z}_i} = \begin{cases} \mathbf{0}, & \boldsymbol{z}_i < \boldsymbol{z}_{ref} \\ \mathbf{1}, & \boldsymbol{z}_i > \boldsymbol{z}_{ref} \end{cases}$$

This ideal rule captures the design intent, but it introduces a discontinuity at the reference thickness, which can create numerical instability during optimization.

Penalization alternatives

To overcome that discontinuity, the paper proposes three penalization families:

- PP — Power Penalization
- MPP — Modified Power Penalization
- EP — Exponential Penalization

These rules smooth or modify the transition below the reference thickness. Their purpose is the same: reduce the structural effectiveness of very thin regions and promote a cleaner separation between main members and noncritical zones. The paper identifies MPP as especially attractive

because it preserves the practical intent of the method while improving continuity at the derivative level.

Stiffness formulation

The global stiffness matrix is assembled from the element stiffness matrices, but each element uses the penalized thickness rather than the raw thickness:

$$\mathbf{K}(\boldsymbol{z}) = \mathcal{A}_{i=1}^{N_e}\mathbf{K}_i(\boldsymbol{z}_i) \quad \text{with} \quad \mathbf{K}_i(\boldsymbol{z}_i) = \boldsymbol{z}_i^*(\boldsymbol{z}_i)\mathbf{K}_u$$

where $\mathbf{K}_u$is the unit-thickness stiffness matrix for a rectangular plane-stress element. This means that two elements with the same geometric area may contribute very differently to the structure depending on their penalized thickness.

Sensitivity analysis

To update the design efficiently, the method uses sensitivities of the objective and constraints. For compliance, the derivative is

$$\frac{\partial \boldsymbol{J}}{\partial \boldsymbol{z}} = -\mathbf{u}^T \frac{\partial \mathbf{K}}{\partial \boldsymbol{z}} \mathbf{u}$$

and for the volume constraint,

$$\frac{\partial \boldsymbol{g}_1}{\partial \boldsymbol{z}} = \frac{\partial \boldsymbol{V}}{\partial \boldsymbol{z}}$$

The derivative of stiffness depends directly on the derivative of the penalized thickness:

$$\frac{\partial \mathrm{K}}{\partial z_i} = \frac{dz_i^*}{dz_i}\mathrm{K}_u$$

These expressions provide the information needed to decide where thickness should increase and where it should decrease during the optimization process.

Structural assumptions behind the method

The formulation uses rectangular plane-stress finite elements and is intended for domains loaded in plane. This is what makes the method computationally efficient: it captures thickness variation without resorting to a full 3D discretization. At the same time, the paper makes clear that the maximum thickness should remain within a range where the plane-stress assumption remains valid.

Practical advantages

The paper highlights several advantages of this formulation:

- it controls minimum thickness,

- it promotes void regions more effectively than classical thickness methods,
- it improves structural performance by allowing members of different thicknesses,
- and it produces layouts with clearer and more practical structural members.

It also shows that allowing a range of thicknesses can outperform traditional fixed-thickness 2D density designs, while still remaining far more efficient than a full 3D density approach.